



TECH HACKATHON On DISABILITY INCLUSION

5th & 6th March 2026

(8 am to 6 pm on both days)

- Prize Pool of Rs. 50000
- Teams of 4 students each

TechTitans

TITLE PAGE

- Problem Statement Title:
AI-Based Gesture & Face Recognition Safety System for Disabled Individuals
- Theme:
AI-Powered Assistive Safety System
- PS Category- Software/Hardware: Software
- Team Name (Registered on portal): TechTitans



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IDEA TITLE

- **Proposed Solution (Describe your Idea/Solution/Prototype):**
 - SafeHomeCam is an AI-based smart safety system that uses hand gestures and face recognition to trigger emergency alerts and protect disabled and elderly individuals.
- **Detailed explanation of the proposed solution**
 - Detects Help, Call, Danger gestures
 - Recognizes authorized/unauthorized faces
 - Activates SafeHouse Mode
 - Sends SMS/Call alerts automatically



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IDEA TITLE

- How it addresses the problem
 - No need to reach phone physically
 - Silent alert for speech-impaired users
 - Intruder detection via face recognition
 - Real-time automated emergency response
- Innovation and uniqueness of the solution
 - Combines gesture + face recognition + alert system
 - Uses AI-based image enhancement for accuracy
 - Designed for disability inclusion
 - Low-cost and scalable solution

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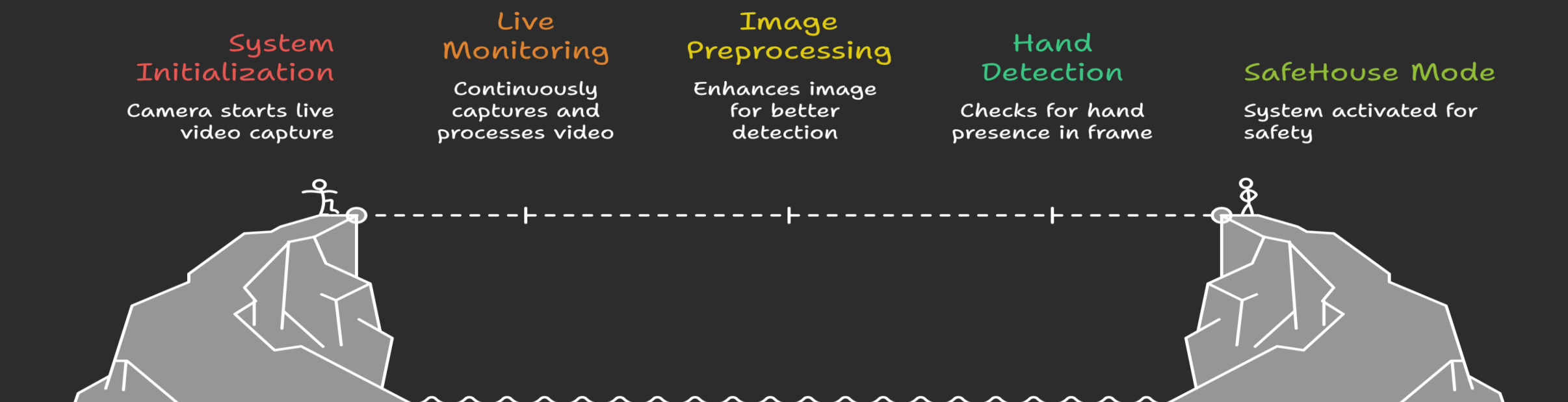
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TECHNICAL APPROACH

- Technology to be used (e.g. programming languages, frameworks, hardware)
 - Python , OpenCV & cvzone
 - TensorFlow (CNN model)
 - face_recognition library & Twilio API for alerts
- Methodology and process for implementation (Flow Charts/Images/working prototype)

Bridging the Path to Accessible Safety





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FEASIBILITY & VIABILITY

- **Analysis of the feasibility of the idea**
 - Uses existing AI frameworks
 - Runs on normal webcam
 - Real-time performance
 - Modular architecture
 - Scalable to cloud deployment
- **Potential challenges and risks**
 - Data privacy concerns
 - Face recognition accuracy
 - Twilio API dependency
 - False positives in low light
- **Strategies for overcoming these challenges**
 - Use encrypted storage
 - Improve dataset quality
 - Use environment variables for API keys
 - Upgrade CNN model



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IMPACT & BENEFITS

- Potential impact on the target audience
 - Improves home security
 - Enables silent emergency help
 - Reduces response time
 - Protects families
- Benefits of the solution (social, economic, environmental, etc.)
 - **Social:**
Safer households
 - **Economic:**
Low-cost smart security alternative
 - **Technological:**
Practical AI + IoT integration

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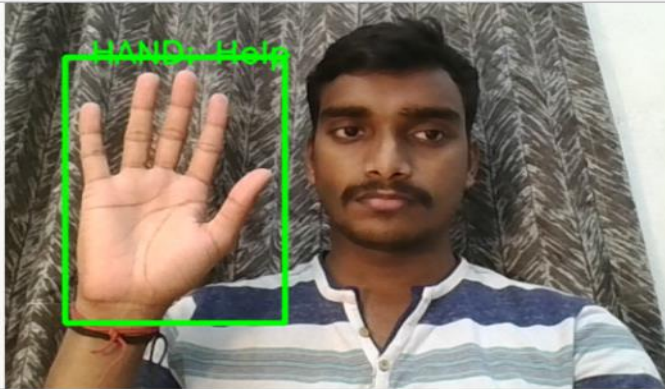
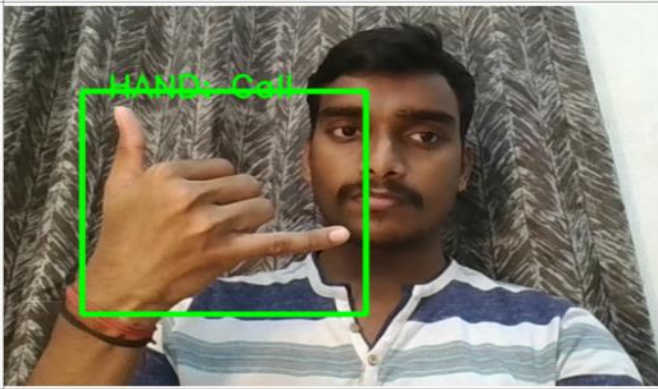
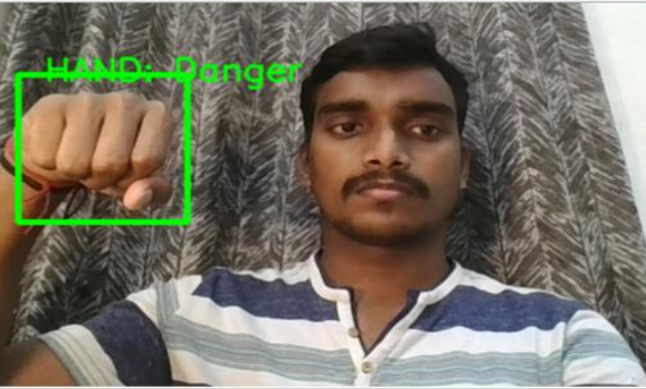
RESEARCH & REFERENCES

- OpenCV Documentation: <https://docs.opencv.org>
- cvzone Library: <https://github.com/cvzone>
- Twilio API Documentation: <https://www.twilio.com/docs>
- face_recognition Library: https://github.com/ageitgey/face_recognition

Demo Video :- <https://www.youtube.com/watch?v=MvPr1x2mDaw&t=141s>

SafeHomeCam

ALL GESTURE CONTROLS

HELP	CALL	DANGER
		
FACE OF AUTHORIZED PERSON	SAFEMODE SYSTEM ON	SAFEMODE SYSTEM OFF
